

Numerical Simulation of Financial SDEs

Problem 4 GBM Benchmark and Non-Affine Volatility

Coursework Report

September 2026

Abstract

This report asks three questions: which method follows a given random path more accurately, whether more accurate paths also give a more accurate mean, and how to assess accuracy without an exact solution. Geometric Brownian motion (GBM) provides an exact benchmark for Euler–Maruyama (EM) and Milstein. Their measured terminal strong orders are 0.500 and 0.991, while both have the same first-order bias in the mean. A non-affine stochastic-volatility model then requires a numerical reference. Three nested reference grids give empirical strong slopes close to one half, but fine-grid call-payoff differences cannot be distinguished reliably from sampling noise. One sample-size study, using existing paired terminal samples, illustrates how additional paths reduce this uncertainty without removing time-discretisation bias.

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Reproducibility: complete Python source, seeds, numerical tables, figures, and run logs accompany this report. Reported confidence intervals are pointwise Monte Carlo intervals.

1 Introduction and theoretical background

The problem pack uses strong and weak convergence, sampling uncertainty and nested grids to assess SDE methods [1]. GBM is a useful starting point because its exact solution lets us check the implementation. The non-affine model then shows how this approach changes when an exact sampler is unavailable.

The results follow three questions. First, how accurately does each method follow the same random path (Section 4)? Second, does a better path approximation also give a better mean (Section 5)? Third, how can a calculation be assessed without an exact solution (Section 6)? Throughout, increasing the number of time steps N improves time resolution, while increasing the number of paths M improves statistical precision.

1.1 GBM and its exact transition

For $S_0 > 0$, the Itô SDE and log transformation are

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (1)$$

$$dX_t = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t, \quad X_t = \log S_t. \quad (2)$$

Itô's correction is essential: omitting $-\sigma^2/2$ changes the mean. Integration gives

$$S_t = S_0 \exp\{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\}. \quad (3)$$

Thus S_T is lognormal, with

$$m = \mathbb{E}[S_T] = S_0 e^{\mu T}, \quad (4)$$

$$v = \text{Var}(S_T) = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1), \quad (5)$$

$$q_p = S_0 \exp\{(\mu - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\Phi^{-1}(p)\}. \quad (6)$$

For the chosen GBM baseline $S_0 = 100$, $\mu = 0.05$, $\sigma = 0.30$, $T = 1$, $m = 105.1271$ and $v = 1040.7868$ (rounded). GBM parameters are experiment choices; the non-affine parameters on page 4 are prescribed by the problem.

1.2 Three methods on a common Brownian path

Let $h = T/N$, $\Delta W_n \sim N(0, h)$. Exact simulation uses

$$S_{n+1} = S_n \exp\{(\mu - \frac{1}{2}\sigma^2)h + \sigma\Delta W_n\}. \quad (7)$$

The price-based EM and scalar Milstein updates are

$$S_{n+1}^E = S_n^E (1 + \mu h + \sigma \Delta W_n), \quad (8)$$

$$S_{n+1}^M = S_n^M \{1 + \mu h + \sigma \Delta W_n + \frac{1}{2}\sigma^2[(\Delta W_n)^2 - h]\}. \quad (9)$$

The extra term follows from $\frac{1}{2}b(S)b'(S)[(\Delta W)^2 - h]$ with $b(S) = \sigma S$ [2]. Under the usual regularity assumptions, terminal strong orders are 1/2 and 1; weak order is 1 for suitable smooth test functions. Applying EM to the GBM log price instead is exact at grid points. Its additive-noise Milstein correction vanishes, so it cannot demonstrate a nontrivial order-one comparison.

2 Monte Carlo methodology

2.1 Coupling and uncertainty

Generate independent paths on the finest grid. A coarse Brownian increment is the sum of the corresponding fine increments, so

$$\Delta W_n^{(h)} = \sum_{j=nr}^{(n+1)r-1} \Delta W_j^{(\delta)}, \quad h = r\delta. \quad (10)$$

For GBM, the exact terminal value uses the same total W_T . For non-affine volatility, coarse and reference states use the same paired Brownian drivers. Independent random numbers across resolutions would introduce a nonvanishing path mismatch and invalidate a strong-order experiment.

For M independent paths, define $A_i(h) = |S_{T,i}^{(h)} - S_{T,i}^{\text{ref}}|$. The terminal L^1 strong-error estimator and its approximate 95% interval are

$$\hat{e}_s(h) = M^{-1} \sum_i A_i(h), \quad \hat{e}_s(h) \pm 1.96 s_A / \sqrt{M}. \quad (11)$$

The observed order is the least-squares slope of $\log \hat{e}_s$ against $\log h$ over an explicitly reported window. Points at different h are correlated because of coupling; the slope is descriptive, and no independent-regression confidence interval is claimed.

For a payoff φ , use signed paired differences

$$D_i(h) = \varphi(S_{T,i}^{(h)}) - \varphi(S_{T,i}^{\text{ref}}), \quad \hat{b}(h) = \bar{D}, \quad \text{CI}_{95} = \bar{D} \pm 1.96 s_D / \sqrt{M}. \quad (12)$$

The weak error is $|\mathbb{E}[D]|$, not $\mathbb{E}|D|$. Signed intervals are retained, including those crossing zero. A log-log trend in an absolute noisy estimate is not sufficient evidence of an order. These intervals quantify sampling error, not the time bias of a numerical reference.

2.2 Resolving GBM weak bias

For both schemes, the conditional expected multiplier is $1 + \mu h$. Consequently

$$\mathbb{E}[S_N^E] = \mathbb{E}[S_N^M] = S_0(1 + \mu h)^{T/h}, \quad (13)$$

$$b_{\text{exact}}(h) = S_0(1 + \mu h)^{T/h} - S_0 e^{\mu T}, \quad (14)$$

$$b_{\text{exact}}(h) = -\frac{1}{2} S_0 e^{\mu T} \mu^2 T h + O(h^2). \quad (15)$$

This identity proves first-order weak bias for the mean in this nondegenerate baseline. It is a validation target, not a Monte Carlo estimate.

The simulation also forms zero-mean control variates C_i and estimates $\overline{D - \beta^\top C}$. The coefficients β are fitted on a separate pilot sample and then held fixed, so the production estimate remains unbiased. Both raw and controlled intervals are retained in the CSV. The pilot is excluded from reported production sample sizes. A short explanation is given on page 6; the five implemented controls and their coefficients are documented in the supplied `gbm.py` and saved results.

3 Model and experimental protocol

3.1 Non-affine stochastic volatility

Write $X = \log S$, and use the prescribed system

$$dX_t = \{\mu - \frac{1}{2}g(Y_t)^2\} dt + g(Y_t) dW_t^{(1)}, \quad (16)$$

$$dY_t = \kappa(\theta - Y_t) dt + \xi \sqrt{1 + Y_t^2} dW_t^{(2)}, \quad (17)$$

$$g(y) = \sigma_{\min} + \frac{\sigma_{\max} - \sigma_{\min}}{1 + e^{-y}}, \quad d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt. \quad (18)$$

Table 1: Prescribed non-affine benchmark.

S_0	Y_0	μ	κ	θ	ξ
100	0	0.05	2	-0.2	0.6
ρ	$\sigma_{\min}$	$\sigma_{\max}$	T	X_0	strike K
-0.7	0.10	0.50	1	$\log 100$	100

From independent standard normals Z_1, Z_2 , form

$$\Delta W^{(1)} = \sqrt{h} Z_1, \quad \Delta W^{(2)} = \sqrt{h}(\rho Z_1 + \sqrt{1 - \rho^2} Z_2). \quad (19)$$

The implemented EM step is simultaneous at the left endpoint:

$$X_{n+1} = X_n + [\mu - \frac{1}{2}g(Y_n)^2]h + g(Y_n)\Delta W_n^{(1)}, \quad (20)$$

$$Y_{n+1} = Y_n + \kappa(\theta - Y_n)h + \xi \sqrt{1 + Y_n^2} \Delta W_n^{(2)}, \quad (21)$$

$$S_{n+1} = \exp(X_{n+1}). \quad (22)$$

The use of Y_n in both updates matters. Replacing it by Y_{n+1} in the first equation changes the discretisation. The logistic implementation avoids exponential overflow, and finiteness and positive prices are checked during simulation.

3.2 Grids and reproducibility

Table 2: Computational protocol; grids are nested powers of two.

Quantity	Value	Quantity	Value
GBM production paths	160,000	Independent pilot	20,000
GBM seed	2026090804	Pilot seed	2026090805
GBM grids N	8 to 1024	GBM batch	2000
Non-affine paths	100,000	Seed	2026090804
Coarse N	8,16,32,64,128,256	Reference N	2048,4096,8192
GBM elapsed	30.9 s	Non-affine elapsed	107.8 s

All calculations use double precision and fixed NumPy generators. Production paths are processed in batches to bound memory. Seeds, batch sizes, runtime versions, elapsed times, and full precision results are stored in the accompanying JSON files. No external market data are used. GBM compares exact, EM, and Milstein on price; the non-affine experiment uses EM with successively finer references. The payoff is $\varphi(S) = (S - 100)^+$, an *undiscounted expected call payoff* under the stated model drift. No risk-neutral interpretation or option price is inferred without a pricing measure and discount rate.

4 Which method follows the same random path more accurately?

4.1 GBM strong convergence

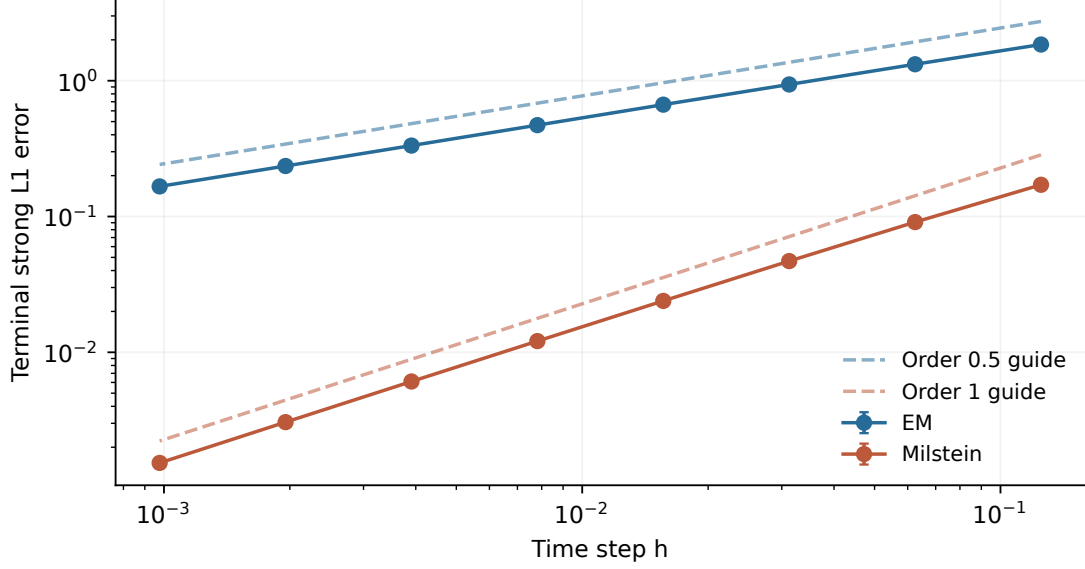


Figure 1: Coupled GBM terminal L^1 error. Error bars are pointwise 95% Monte Carlo intervals; many are smaller than the marker. Reference lines indicate orders $1/2$ and 1 .

Table 3: GBM strong error estimates with 95% interval half-widths.

h	EM	Milstein
1/8	1.84750 ± 0.00809	0.17127 ± 0.00100
1/16	1.32218 ± 0.00562	0.09096 ± 0.00048
1/32	0.93847 ± 0.00396	0.04697 ± 0.00023
1/64	0.66641 ± 0.00278	0.02391 ± 0.00011
1/128	0.47071 ± 0.00196	0.01209 ± 0.00006
1/256	0.33297 ± 0.00139	0.00610 ± 0.00003
1/512	0.23550 ± 0.00098	0.00306 ± 0.00001
1/1024	0.16658 ± 0.00069	0.00153 ± 0.00001

The fitted slopes are 0.500 for EM and 0.991 for Milstein on $N = 64, 128, 256, 512, 1024$. The observed rates match the theoretical distinction: EM omits the quadratic stochastic increment, whereas Milstein accounts for it. At the finest step, Milstein has a substantially lower pathwise error. This comparison uses exactly the same Brownian paths, so differences between methods arise from the update rules rather than from different realizations of the driving noise.

A half-order method reduces error by about $1/\sqrt{2}$ when the step is halved; a first-order method approximately halves the error. From $N = 64$ to 1024 , the step decreases by a factor of 16 . EM's error falls from 0.66641 to 0.16658 , a factor of 4.00 , close to $\sqrt{16}$. Milstein's error falls from 0.02391 to 0.00153 , a factor of about 15.6 , close to 16 . These are terminal price errors, not maximum errors along the whole path. The exact grid values have no time-discretisation error.

5 Does a better path approximation give a better mean?

5.1 GBM weak convergence

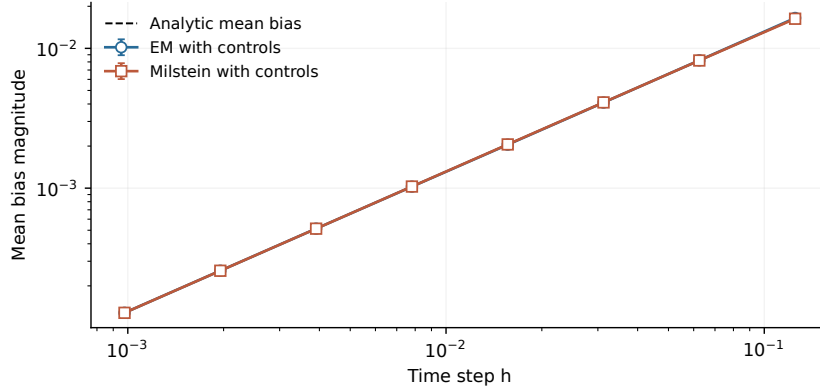


Figure 2: GBM mean bias magnitude. Controlled paired estimates are compared with the analytic discrete-mean bias. Raw estimates and intervals remain available in the saved CSV data.

Table 4: Signed weak bias; controlled estimates plus 95% half-widths.

h	Analytic bias	EM controlled	Milstein controlled
1/8	-1.636×10^{-2}	$-1.649 \times 10^{-2} \pm 1.940 \times 10^{-4}$	$-1.629 \times 10^{-2} \pm 9.901 \times 10^{-5}$
1/32	-4.102×10^{-3}	$-4.096 \times 10^{-3} \pm 2.811 \times 10^{-5}$	$-4.106 \times 10^{-3} \pm 1.399 \times 10^{-5}$
1/128	-1.026×10^{-3}	$-1.025 \times 10^{-3} \pm 3.653 \times 10^{-6}$	$-1.027 \times 10^{-3} \pm 1.810 \times 10^{-6}$
1/512	-2.566×10^{-4}	$-2.567 \times 10^{-4} \pm 4.660 \times 10^{-7}$	$-2.566 \times 10^{-4} \pm 2.301 \times 10^{-7}$
1/1024	-1.283×10^{-4}	$-1.284 \times 10^{-4} \pm 1.649 \times 10^{-7}$	$-1.283 \times 10^{-4} \pm 8.144 \times 10^{-8}$

For the mean, the answer is no: EM and Milstein have identical analytic biases despite their different strong errors. Each step has the same expected multiplier $1 + \mu h$, so equation (14) applies to both. The analytic fitted order is 0.9993 and the controlled Monte Carlo slopes are 0.999 and 1.000. This conclusion concerns the mean observable; it does not say that every payoff has the same bias for both methods.

The purpose of a control variate is to remove sampling noise without changing the quantity being estimated. If $D = S_N^h - S_T$ and a random variable C has known mean zero, then, for a fixed coefficient β ,

$$\mathbb{E}[D - \beta C] = \mathbb{E}[D]. \quad (23)$$

When C varies with D , subtracting a suitable multiple can make the sample average much less noisy. The implementation uses five such variables calculated from the same Brownian increments. Their coefficients are fitted on 20,000 separate pilot paths and then held fixed for the 160,000 production paths. The detailed formulas remain in `gbm.py`; they are a computational aid, rather than an additional convergence method.

Table 5: GBM at $N = 1024$: standard errors of the paired mean-bias estimates.

Method	Before controls	After controls
EM	5.464e-04	8.415e-08
Milstein	5.258e-06	4.155e-08

Table 5 explains the need for variance reduction: EM's raw standard error exceeds the mean-bias magnitude 1.28×10^{-4} . Milstein's raw interval already excludes zero. The controls reduce estimation noise; they change neither the time-stepping methods nor their bias. The analytic bias is only a validation target.

5.2 Distribution and positivity

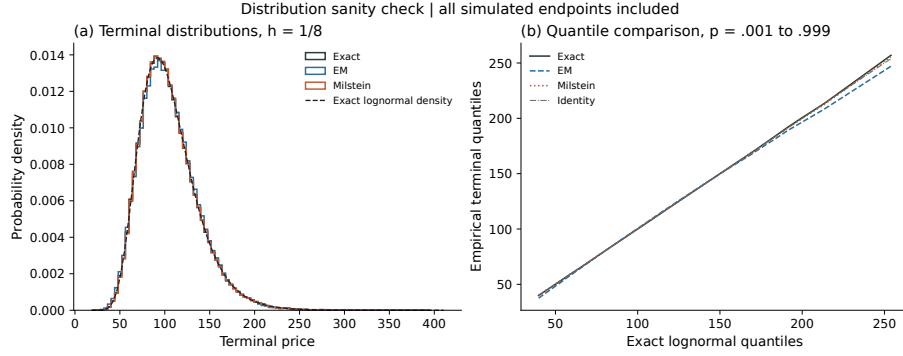


Figure 3: GBM terminal distribution compared with the exact lognormal law. The histogram is a sampling diagnostic; quantiles give a more precise distributional comparison.

Table 6: Terminal quantiles; numerical columns use $h = 1/8$.

p	Lognormal	Exact simulation	EM	Milstein
0.01	50.012	49.928	48.387	50.213
0.05	61.357	61.378	60.512	61.615
0.25	82.091	82.151	82.303	82.263
0.50	100.501	100.557	101.088	100.582
0.75	123.041	123.164	123.653	123.057
0.95	164.618	164.485	163.370	164.043
0.99	201.961	202.402	198.376	201.539

The exact-simulation mean is 105.1987 and its variance is 1045.0120; the mean interval is [105.0403, 105.3571]. The analytic mean is 105.1271 and variance is 1040.7868. Sampling fluctuations in empirical moments and quantiles remain even for exact simulation; they must not be described as discretisation error.

For $S_n > 0$, EM becomes negative with one-step probability

$$\mathbb{P}(S_{n+1}^E < 0 \mid S_n > 0) = \Phi\left(-\frac{1 + \mu h}{\sigma\sqrt{h}}\right). \quad (24)$$

For Milstein, completing the square gives the multiplier

$$\frac{1}{2}\sigma^2(\Delta W + 1/\sigma)^2 + \frac{1}{2} + (\mu - \frac{1}{2}\sigma^2)h. \quad (25)$$

It can be negative only if $(\sigma^2 - 2\mu)h > 1$. Here $\sigma^2 - 2\mu = -0.01$, so Milstein is positive for every $h > 0$ in this parameter set. EM has a nonzero theoretical probability of a negative update, although that probability is extremely small for these parameters.

Table 7: Observed GBM negative terminal frequency: 160,000 paths for each method at each grid.

N	EM: $S_N < 0$	Milstein: $S_N < 0$
8, 16, 32, 64	0/160,000 (0%) at each grid	0/160,000 (0%) at each grid
128, 256, 512, 1024	0/160,000 (0%) at each grid	0/160,000 (0%) at each grid

The saved counts are for $S_N \leq 0$ and are all zero, so the strictly negative counts in Table 7 are also zero. These are terminal frequencies, not counts of paths ever becoming negative: intermediate GBM prices were not recorded by this diagnostic. Zero observed negatives does not prove that EM preserves positivity. Negative prices could distort payoff estimates; clipping them to zero would also change the method. Neither scheme is clipped here.

6 How can we assess accuracy without an exact solution?

6.1 Strong differences against a numerical reference

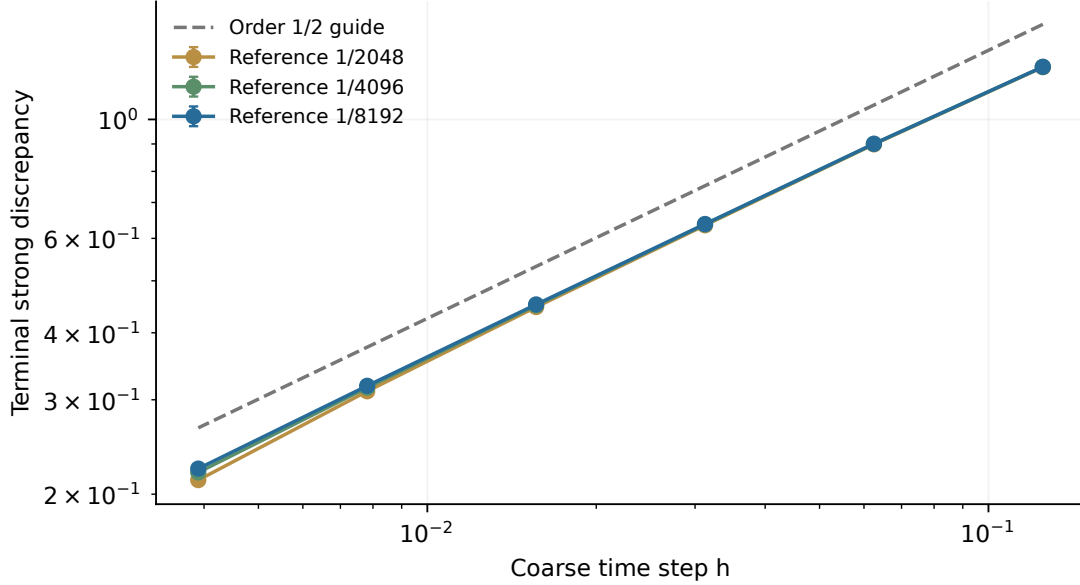


Figure 4: Terminal strong differences against three coupled EM reference grids. The reference is numerical, so these are not exact-error curves.

Table 8: Strong discrepancy against $h_{\text{ref}} = 1/8192$; 95% half-widths.

h	$\mathbb{E} S_h - S_{\text{ref}} $	Last reference difference / error
1/8	1.25354 ± 0.00702	3.2%
1/16	0.90087 ± 0.00489	4.5%
1/32	0.63818 ± 0.00341	6.3%
1/64	0.45176 ± 0.00241	8.9%
1/128	0.31854 ± 0.00170	12.6%
1/256	0.22323 ± 0.00119	18.0%

Against the finest reference, the empirical slope is 0.491 over $N = 8, 16, 32, 64$. The result is consistent with approximately half-order strong behavior on this window. Curves bend when a coarse grid approaches a reference grid: two nearly identical discretisations can agree well while both retain a nonzero error relative to the continuous SDE.

If S^* denotes the unknown exact terminal value, the reverse triangle inequality yields

$$|\mathbb{E}|S_h - S^*| - \mathbb{E}|S_h - S_r|| \leq \mathbb{E}|S_r - S^*|. \quad (26)$$

The right-hand side is unknown. Replacing r by a finer grid measures sensitivity of the benchmark, not that unknown error itself. We therefore compare the fitted slope across references as well as reporting the differences between reference solutions.

6.2 Weak payoff differences

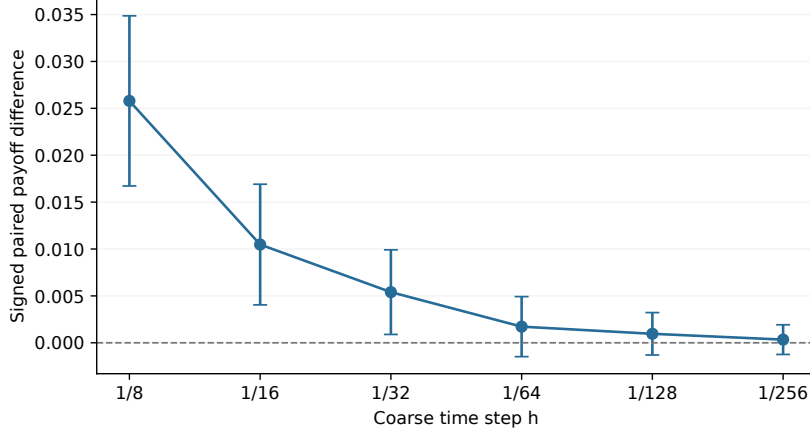


Figure 5: Signed paired differences in the expected call payoff relative to the finest EM reference. Error bars are 95% sampling intervals; the zero line identifies unresolved bias signs.

Table 9: Signed paired payoff differences against $h_{\text{ref}} = 1/8192$.

h	Estimated bias	95% interval	Zero excluded
1/8	0.02579	[0.01672, 0.03486]	Yes
1/16	0.01048	[0.00405, 0.01691]	Yes
1/32	0.00540	[0.00089, 0.00991]	Yes
1/64	0.00172	[-0.00148, 0.00492]	No
1/128	0.00096	[-0.00130, 0.00322]	No
1/256	0.00034	[-0.00124, 0.00192]	No

At $N = 8, 16, 32$, the intervals exclude zero, so these coarse-grid payoff differences are distinguishable from sampling noise. At $N = 64, 128, 256$, the intervals include zero: the available sample cannot reliably resolve the sign. This does not prove zero bias, and no weak order is assigned to the complete grid range. The payoff has a corner at the strike, so a weak order should not simply be assumed from smooth-function theory. The paired intervals show whether the estimated difference can be distinguished from sampling noise; they do not include reference bias.

6.3 Why use a call payoff instead of just the mean?

The continuous model satisfies $\mathbb{E}[S_T] = 100e^{0.05}$. However, the log-EM method also gives this mean at *every* step size. To see why, set $g_n = g(Y_n)$. At the start of the step, S_n and g_n are known and the next increment is $N(0, h)$. Hence

$$\begin{aligned} \mathbb{E}[S_{n+1} \mid \mathcal{F}_{t_n}] &= S_n e^{(\mu - g_n^2/2)h} \mathbb{E}[e^{g_n \Delta W_n^{(1)}} \mid \mathcal{F}_{t_n}] \\ &= S_n e^{(\mu - g_n^2/2)h} e^{g_n^2 h/2} = S_n e^{\mu h}. \end{aligned} \tag{27}$$

Here $\mathcal{F}_{t_n}$ means the information available at the start of the step. Repeating this calculation gives $\mathbb{E}[S_N] = 100e^{0.05}$. Thus the mean is a useful implementation check, but has zero time-stepping bias for this scheme and cannot reveal its weak convergence rate.

Equal means do not imply equal expected payoffs. For example, a price always equal to 100 and a price equally likely to be 80 or 120 both have mean 100. Their expected call payoffs $(S - 100)^+$ are respectively 0 and 10. The call payoff therefore tests more of the terminal distribution than its mean. Unlike the mean, its expectation is not automatically preserved by the log-EM update, so changing the step size can produce a genuine weak bias. This makes it a nontrivial weak observable; it is not the only possible choice, and does not guarantee a measurable bias at every grid.

6.4 Does the answer change when the reference is refined?

Table 10: Coupled refinement of the numerical reference.

Refinement $N_a \rightarrow N_b$	Strong difference	Signed payoff difference and 95% CI
2048 $\rightarrow$ 4096	0.05679 ± 0.00030	-0.000023 [-0.000423, 0.000377]
4096 $\rightarrow$ 8192	0.04016 ± 0.00021	0.000094 [-0.000188, 0.000376]

The average absolute difference falls from 0.05679 for 2048 versus 4096 steps to 0.04016 for 4096 versus 8192 steps. The second difference is about 0.707 of the first, consistent with half-order scaling. Relative to the coarse-to-fine differences, 0.04016 represents 3.2% to 8.9% on the $N = 8, 16, 32, 64$ window, but 12.6% and 18.0% at $N = 128, 256$. Reference resolution therefore matters more in relative terms on the finer coarse grids. The principal slope uses the less reference-sensitive four-grid window. These comparisons measure sensitivity to reference refinement; they do not bound the unknown error against the exact solution.

Table 11: Same fitting window $N = 8, 16, 32, 64$, different numerical references.

Reference steps	Empirical strong slope
2048	0.496108
4096	0.493008
8192	0.491450

Table 11 keeps the fitting window fixed and changes only the reference. The three slopes, 0.496, 0.493 and 0.491, support a stable half-order trend on this window. Their agreement is numerical evidence, not an exact-error bound.

Table 12: Brownian and state diagnostics at the finest non-affine grid.

Quantity	Observed	Target or scope
Increment variance 1	1.221×10^{-4}	1.221×10^{-4}
Increment variance 2	1.221×10^{-4}	1.221×10^{-4}
Increment correlation	-0.699675	-0.700000
Sampled increments	4,194,304	Predetermined first batch
Total path updates	1,484,000,000	All grids
Finite intermediate states	Pass	X and Y
Positive finite prices	Pass	All observed states

The variance targets for both Brownian increments are h , and their target correlation is -0.7 . The diagnostics are calculated from the generated fine increments, not inferred from the generator formula. State checks include intermediate steps, because a finite terminal value alone would not rule out an invalid intermediate update.

The finest-grid mean is 105.08796 with 95% interval [104.89962, 105.27629]. The corresponding undiscounted payoff estimate is 14.53804 with interval [14.40515, 14.67093]. This latter interval describes sampling uncertainty about the *finest discrete model*; the refinement table is still needed to assess its bias.

As requested in the problem pack, EM is used for this two-noise model. A multidimensional Milstein method would need additional cross-noise terms; the scalar GBM formula cannot simply be copied into this system. No such extension is needed for the present comparison.

6.5 One sample-size study: what do more paths improve?

Fix the coarse grid at $N = 64$ and the reference at $N_{\text{ref}} = 8192$. From the saved terminal arrays, form the paired payoff differences

$$D_i = (S_{T,i}^{(1/64)} - 100)^+ - (S_{T,i}^{(1/8192)} - 100)^+. \quad (28)$$

Use the first 10,000, 40,000 and 100,000 pairs in their saved order. These nested subsets reuse observations, so the three estimates are correlated; they are not independent repeated experiments. No new paths or alternative seeds are generated.

Table 13: One fixed grid pair: $N = 64$ versus $N_{\text{ref}} = 8192$. Nested prefixes of the saved paired sample.

Paths M	Signed payoff difference	Standard error	95% interval
10,000	0.009045	0.005149	[-0.001048, 0.019137]
40,000	0.004942	0.002592	[-0.000139, 0.010022]
100,000	0.001722	0.001633	[-0.001479, 0.004923]

For independent paths within each subset, the standard error is $s_D/\sqrt{M}$. If the sample variance changes little, quadrupling M should halve this error. In Table 13, it falls from 0.005149 at 10,000 paths to 0.002592 at 40,000, a ratio of 0.503. At 100,000 paths it is 0.001633, about 0.317 of its initial value, close to $1/\sqrt{10}$. Thus the observed uncertainty decreases at the expected scale.

All three intervals still contain zero. More paths improve precision but do not guarantee that this bias becomes detectable at the available sample size. The decreasing point estimates do not show that the method's bias is decreasing: both time grids are fixed, so all three estimates target the same unknown expected difference. Nor should the 100,000-path estimate be treated as its exact value.

Practical conclusion. Increasing M reduces sampling uncertainty. Increasing N changes the time discretisation. This study addresses only the first effect; it neither establishes a weak time-convergence order nor removes bias in the numerical reference. It explains why unresolved fine-grid payoff differences are a statistical limitation, not evidence that their true bias is zero.

7 Discussion and conclusion

Path accuracy. GBM supplies a direct answer because the exact and numerical values use the same Brownian path. The measured strong slopes, 0.500 and 0.991, show the expected advantage of Milstein. Refining the step by a factor of 16 reduces EM’s terminal error by about 4 and Milstein’s by about 15.6. This compares accuracy at fixed grids; it is not a claim about speed at equal computing cost.

Mean accuracy. More accurate paths do not automatically give a smaller bias for every observable. Both GBM methods have the same analytic first-order mean bias; controlled estimates recover slopes near one. The histogram and quantiles examine distributional features that the mean alone cannot check. The observed negative terminal frequency is zero on all tested grids, while the EM formula still permits negative updates.

Accuracy without an exact solution. For the non-affine model, the strong slopes remain close to one half across three references. Coarse-grid call-payoff differences are distinguishable from sampling noise; the finer-grid differences are not. No weak order is assigned to the full range. The sample-size study confirms that adding paths narrows intervals approximately as $M^{-1/2}$, but all tested sample sizes still leave the $N = 64$ payoff difference unresolved.

The calculations therefore answer the three questions at different levels: exact-solution validation for GBM, an analytic explanation of its mean bias, and refinement evidence for the non-affine model. Additional paths cannot correct an insufficiently fine reference, and finer steps alone cannot remove sampling uncertainty. These conclusions concern the stated parameters, terminal error measures and grid ranges.

Reproducibility

The accompanying Python files and saved results record the seeds, sample sizes and updates. The sample-size study reads existing paired terminal arrays; it does not rerun the SDE simulation. Its full-sample mean and standard error agree with the saved experiment. The appendix records the core updates and reproduction commands.

AI acknowledgement

OpenAI Codex was used to help explain the mathematical methods, develop and check the Python code, carry out numerical calculations, and draft and revise the report. This assistance included improving the clarity of the discussion and presentation of the results.

References

- [1] *Problem Pack: Five Suggested Directions for the ODE IVP Project*. Supplied coursework document, Direction 4, pp. 6–7.
- [2] D. J. Higham. An algorithmic introduction to numerical simulation of stochastic differential equations. *SIAM Review*, 43(3):525–546, 2001. doi:10.1137/S0036144500378302.
- [3] P. E. Kloeden and E. Platen. *Numerical Solution of Stochastic Differential Equations*. Springer, 1992. doi:10.1007/978-3-662-12616-5.

A Python implementation and reproduction

The following listing states the core mathematical operations. The executable modules additionally implement batching, control-variate pilot fitting, uncertainty estimates, diagnostics, exports, and plots; their complete source is supplied alongside this report.

Listing 1: Core SDE updates and coupled error statistics.

```
1 import numpy as np
2
3 def coarsen(dw_fine, n_coarse):
4     m, n_fine = dw_fine.shape
5     assert n_fine % n_coarse == 0
6     return dw_fine.reshape(m, n_coarse, -1).sum(axis=2)
7
8 def gbm_terminal(dw, s0=100., mu=.05, sigma=.3, T=1.):
9     h = T / dw.shape[1]
10    exact = s0 * np.exp((mu-.5*sigma**2)*T
11                      + sigma*dw.sum(axis=1))
12    em = s0 * np.prod(1 + mu*h + sigma*dw, axis=1)
13    mil = s0 * np.prod(1 + mu*h + sigma*dw
14                      + .5*sigma**2*(dw*dw-h), axis=1)
15    return exact, em, mil
16
17 def log_em_step(x, y, dw1, dw2, h):
18     # Values on both right-hand sides are at the old time.
19     sigmoid = np.exp(-np.logaddexp(0., -y))
20     g = .1 + .4*sigmoid
21     x_new = x + (.05-.5*g*g)*h + g*dw1
22     y_new = y + 2*(-.2-y)*h + .6*np.hypot(1.,y)*dw2
23     return x_new, y_new
24
25 def mean_ci(values):
26     m = len(values)
27     mean = np.mean(values)
28     half_width = 1.96*np.std(values, ddof=1)/np.sqrt(m)
29     return mean, mean-half_width, mean+half_width
30
31 # Brownian drivers from independent standard normals:
32 # dw1 = sqrt(h)*z1
33 # dw2 = sqrt(h)*(rho*z1 + sqrt(1-rho**2)*z2)
34 # Strong: mean_ci(abs(s_coarse-s_reference))
35 # Weak: mean_ci(maximum(s_coarse-100,0)
36 #               -maximum(s_reference-100,0))
```

Run the experiments from the original package directory using Python 3.12:

```
python -m pip install -r requirements.txt
python run_all.py

# To validate supplied results without resimulation:
python verify.py
```

Rebuild the supplementary tables and compile from `undergraduate_revision/`:

```
python sample_size_study.py
tectonic Coursework_Undergraduate.tex
```

In the original package, `results/` contains machine-readable tables and logs; `figures/` contains vector PDFs and PNGs. The supplied PDF is already compiled. The appendix listing is explanatory; the tests exercise the actual imported implementation functions. Full reproduction includes both simulation modules, while rebuilding only the report tables does not resimulate paths.